# Modulo 2

Calculito

2026-04-08

CLASE 1

## R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see http://rmarkdown.rstudio.com.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

## Primeros comandos de R

R es un lenguaje muy parecido a MATLAB y trabaja normalmente con variables que pueden ser matrices

```
A<-49
B=sqrt(2500)
B
```

```
## [1] 50
```

```
C=B-A
C
```

```
## [1] 1
```

```
summary(cars)
```

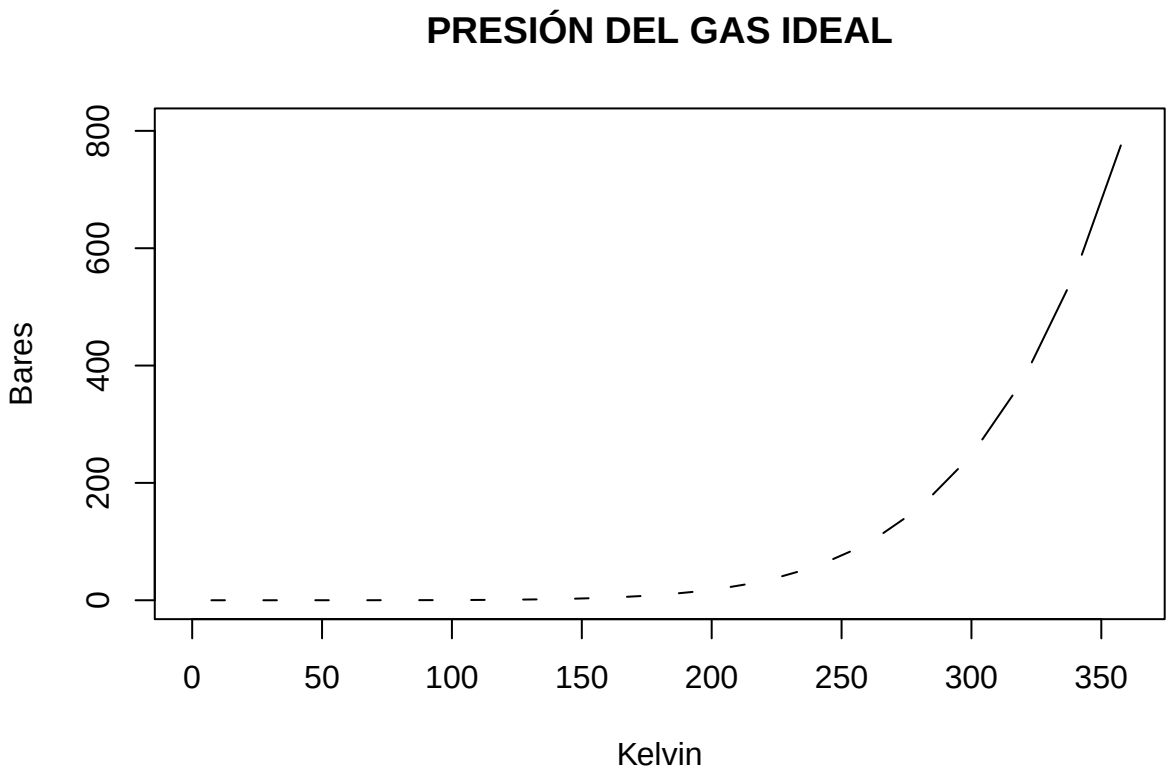
```
##      speed           dist       
##  Min.   : 4.0   Min.   :  2.00  
##  1st Qu.:12.0   1st Qu.: 26.00  
##  Median :15.0   Median : 36.00  
##  Mean   :15.4   Mean   : 42.98  
##  3rd Qu.:19.0   3rd Qu.: 56.00  
##  Max.   :25.0   Max.   :120.00  
```

## Uso de data sets (bases de datos) en R

Escribiendo data() en **"Console"**, se abrirá una pestaña contigua titulada "R data sets", compuesta de una serie de ítems que contienen distintas series de datos. Por ejemplo, de la efectividad de un repelente o de accidentes automovilísticos en Gran Bretaña entre los años 1969 y 1984. Para desplegar los datos, ingresamos el título del data set en la consola. De ser una base de datos con más de una variable, el signo $ a continuación del título.

## Including Plots

You can also embed plots, for example:

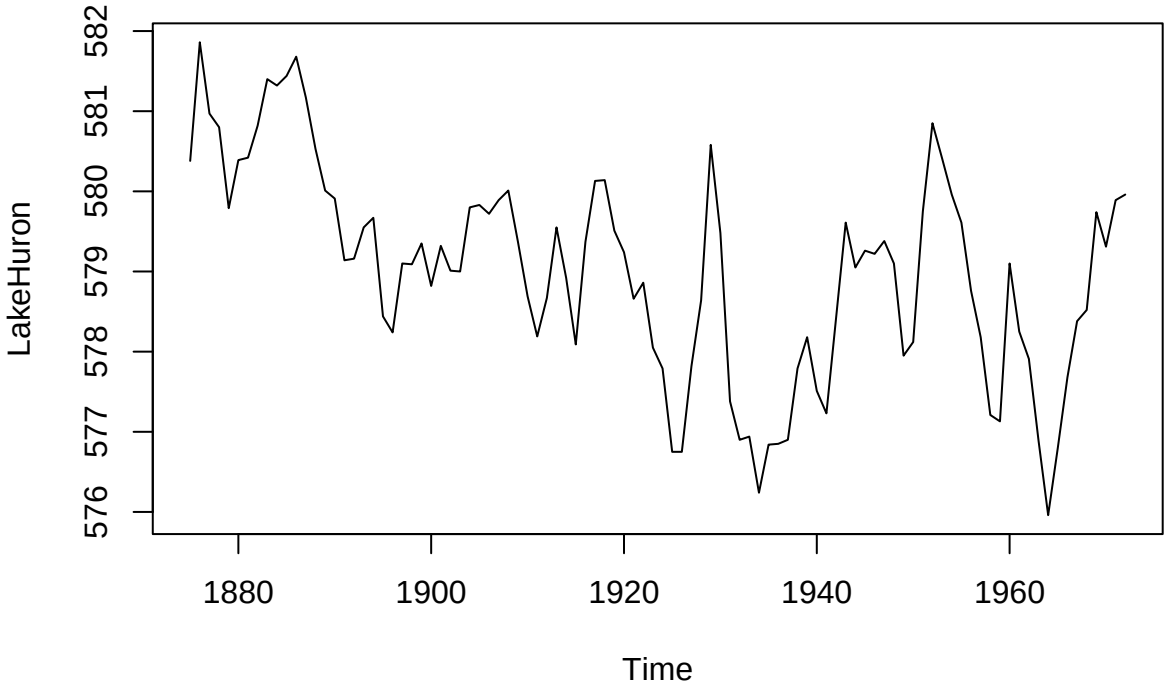



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.

## Explicación del comando plot

Mediante el comando (en lenguaje de programación R) es posible graficar la información de los data sets insertando "chunks" de código en R y escribiendo plot(título del data set). Escribiendo en la consola **?plot** se puede indagar sobre las diversas modificaciones al gráfico, ingresadas luego de una coma después del título del data set. En caso de ser texto, se escribe entre comillas dobles.

### Nivel del lago Huron a lo largo del tiempo



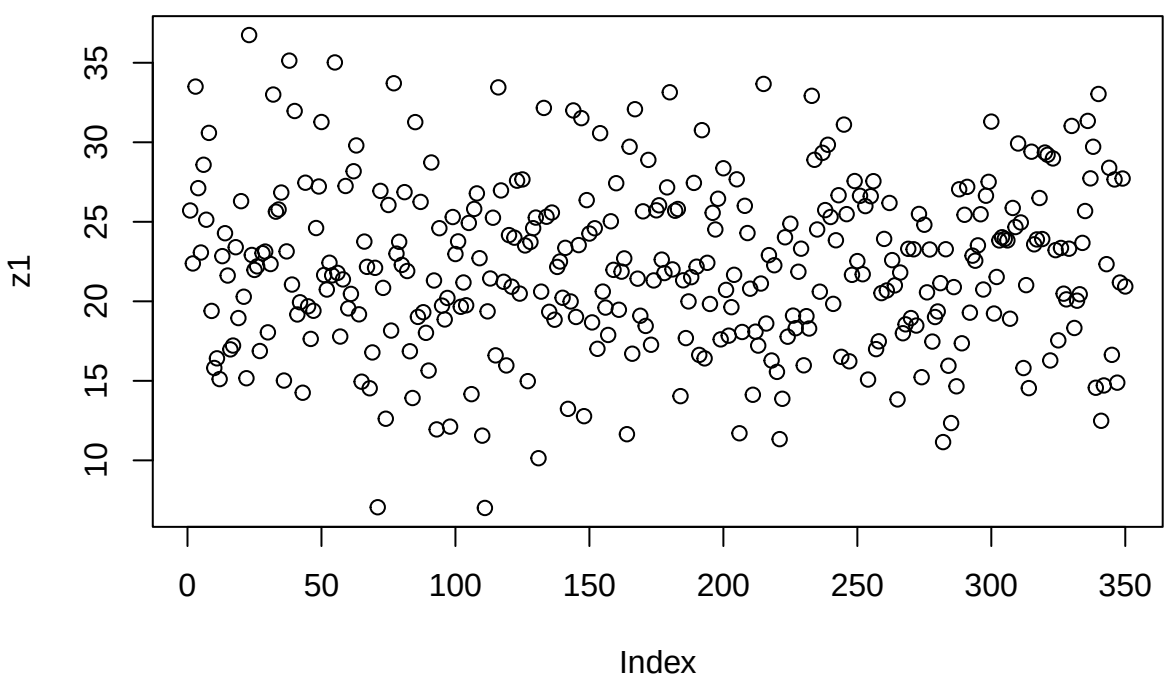

### Funciones estadísticas

```
z1<-rnorm(350,22,5)
z1
```

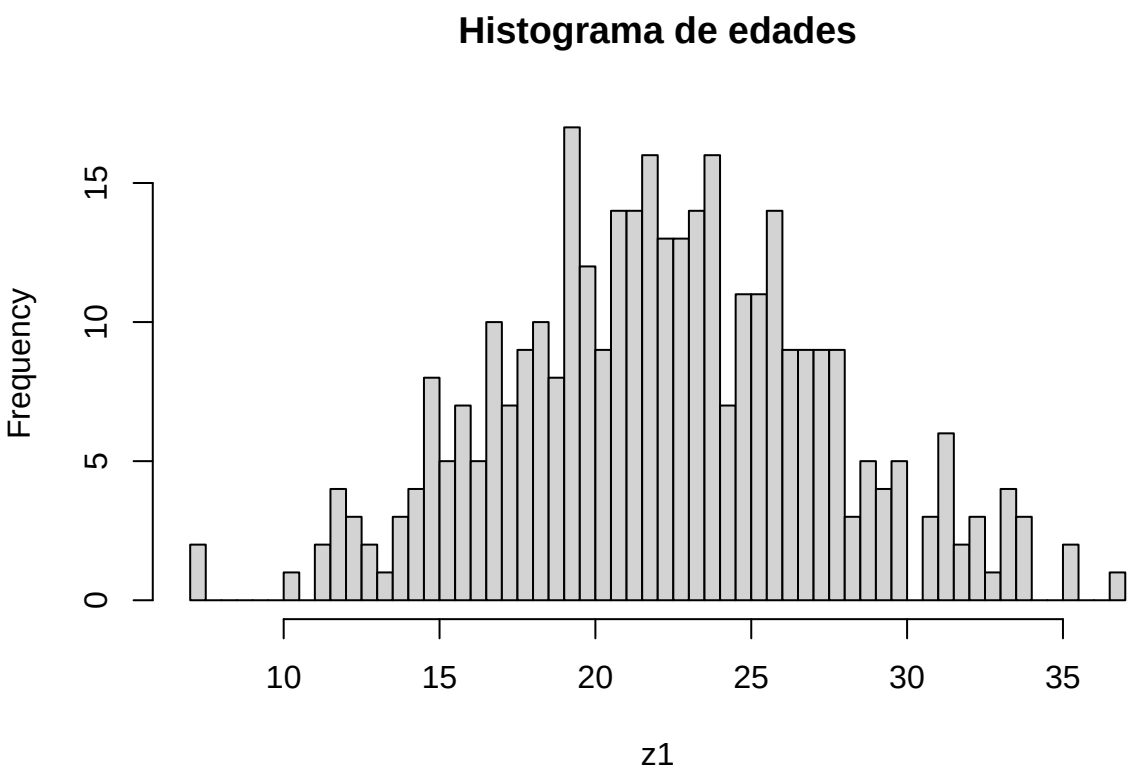
```
##   [1] 25.710167 22.380678 33.500774 27.114968 23.068160 28.586045 25.130479
##   [8] 30.590511 19.396642 15.808606 16.414625 15.109755 22.840724 24.282245
##  [15] 21.622020 16.977849 17.218898 23.402328 18.947665 26.298302 20.287988
##  [22] 15.164039 36.740993 22.913401 21.973814 22.200436 16.873940 23.025445
##  [29] 23.128057 18.047864 22.343656 33.001285 25.637549 25.772471 26.839843
##  [36] 15.018333 23.139223 35.132966 21.053735 31.967013 19.173362 19.939324
##  [43] 14.253111 27.457525 19.685353 17.635412 19.406681 24.607355 27.219315
##  [50] 31.273751 21.655981 20.741032 22.425871 21.633622 35.020014 21.786209
##  [57] 17.785603 21.378515 27.251439 19.555895 20.465393 28.179424 29.800088
##  [64] 19.181955 14.945896 23.755097 22.167502 14.532844 16.786528 22.105808
##  [71]  7.052770 26.942280 20.844429 12.618300 26.055966 18.160166 33.709426
##  [78] 22.998005 23.742475 22.289765 26.865006 21.905290 16.859691 13.919832
##  [85] 31.267601 19.020345 26.250257 19.314519 18.018285 15.646315 28.727978
##  [92] 21.309637 11.946075 24.597074 19.727614 18.860119 20.210671 12.120795
##  [99] 25.299649 22.972765 23.767056 19.657026 21.183290 19.750223 24.927170
## [106] 14.164089 25.807364 26.785702 22.710646 11.553840  7.007907 19.367020
## [113] 21.434634 25.256570 16.600904 33.451578 26.971940 21.227199 15.967842
## [120] 24.163314 20.919126 24.001890 27.586695 20.484169 27.663554 23.512780
## [127] 14.979908 23.734597 24.597511 25.264405 10.136542 20.609363 32.159681
## [134] 25.318371 19.339635 25.582389 18.842208 22.162198 22.503636 20.222949
## [141] 23.361165 13.238384 19.990327 32.002113 19.015270 23.534302 31.516633
## [148] 12.778579 26.362080 24.261180 18.673579 24.593988 17.013282 30.571500
## [155] 20.611326 19.605350 17.885371 25.031283 21.978134 27.425865 19.469072
## [162] 21.865567 22.698297 11.645397 29.718651 16.704526 32.078599 21.424663
## [169] 19.092619 25.657241 18.451751 28.890942 17.267858 21.312059 25.720808
```

```
## [176] 26.037303 22.622137 21.777552 27.164739 33.142993 22.006676 25.695020
## [183] 25.804812 14.038296 21.318915 17.687565 19.993254 21.518456 27.445664
## [190] 22.175609 16.629198 30.760036 16.412225 22.420845 19.833805 25.563225
## [197] 24.516923 26.449626 17.606888 28.365712 20.717608 17.839776 19.630850
## [204] 21.666665 27.671448 11.703993 18.071989 26.004487 24.292990 20.793609
## [211] 14.124036 18.092323 17.216187 21.118319 33.668050 18.605538 22.903601
## [218] 16.278867 22.267940 15.562956 11.334269 13.871551 24.025367 17.774534
## [225] 24.881352 19.101632 18.309202 21.859584 23.309198 15.975971 19.066394
## [232] 18.289132 32.920563 28.897221 24.516768 20.605653 29.334925 25.736461
## [239] 29.843948 25.302178 19.838582 23.837865 26.669215 16.508889 31.111927
## [246] 25.481259 16.227979 21.664761 27.561170 22.528918 26.634084 21.703605
## [253] 25.987842 15.080774 26.610005 27.553915 16.993278 17.484886 20.516678
## [260] 23.922732 20.681896 26.183090 22.589673 21.002836 13.831817 21.811243
## [267] 17.993232 18.553789 23.301993 18.946548 23.273149 18.476648 25.492235
## [274] 15.229189 24.814695 20.563847 23.259378 17.468366 19.013147 19.360908
## [281] 21.143817 11.149538 23.275148 15.949772 12.345954 20.892755 14.656320
## [288] 27.044438 17.360665 25.440426 27.196297 19.285861 22.870329 22.554195
## [295] 23.518040 25.474997 20.742560 26.639846 27.511479 31.299110 19.231971
## [302] 21.530987 23.828103 24.029220 23.921764 23.831606 18.905317 25.869823
## [309] 24.668986 29.924016 24.957323 15.804358 21.017480 14.542670 29.407413
## [316] 23.582506 23.891457 26.494827 23.914632 29.354648 29.211039 16.281334
## [323] 28.983447 23.216950 17.545974 23.354385 20.481893 20.121922 23.307983
## [330] 31.027930 18.323246 20.039422 20.442369 23.674198 25.677837 31.342521
## [337] 27.729033 29.717750 14.566023 33.036499 12.489160 14.710015 22.336364
## [344] 28.394253 16.635167 27.661655 14.886303 21.181856 27.720730 20.932554
```

```r
plot(z1)
```



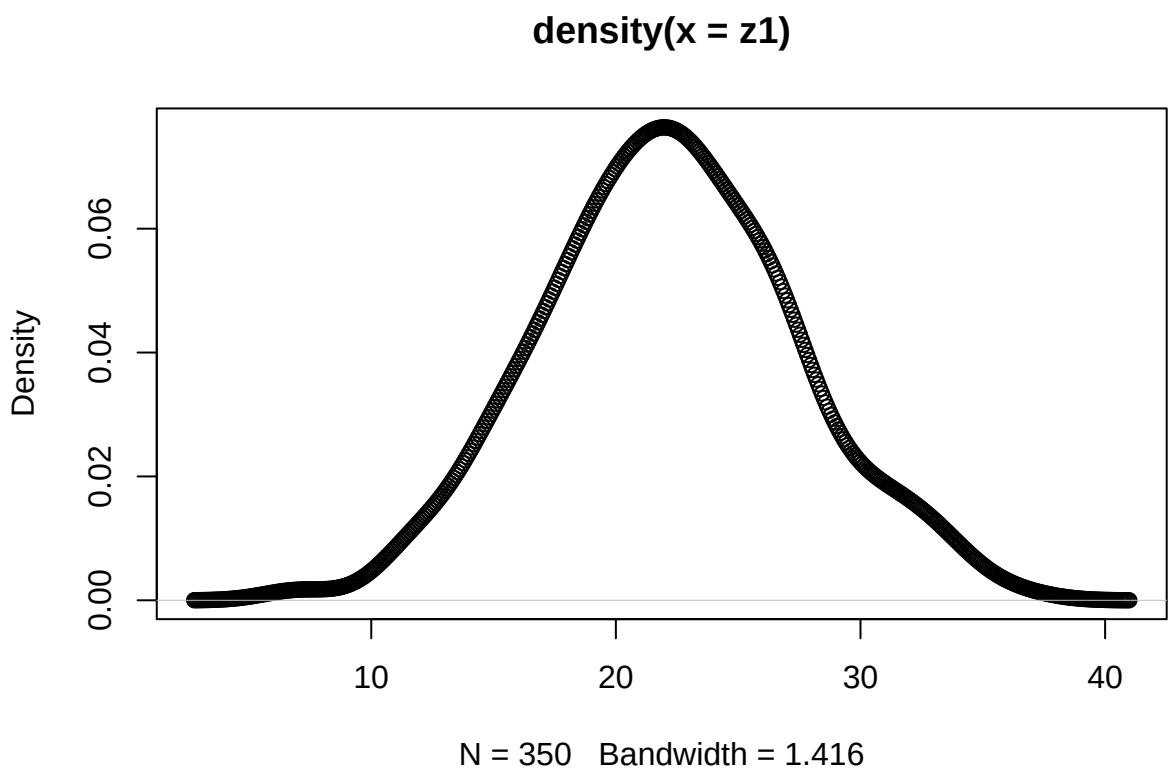

```r
hist(z1,main="Histograma de edades",breaks=50)
```




```r
density(z1)
```

```
##
## Call:
##  density.default(x = z1)
##
## Data: z1 (350 obs.); Bandwidth 'bw' = 1.416
##
##        x                y
##  Min.   : 2.761   Min.   :9.286e-06
##  1st Qu.:12.318   1st Qu.:1.826e-03
##  Median :21.874   Median :1.645e-02
##  Mean   :21.874   Mean   :2.611e-02
##  3rd Qu.:31.431   3rd Qu.:4.920e-02
##  Max.   :40.988   Max.   :7.636e-02
```

```r
plot(density(z1),type="b")
```



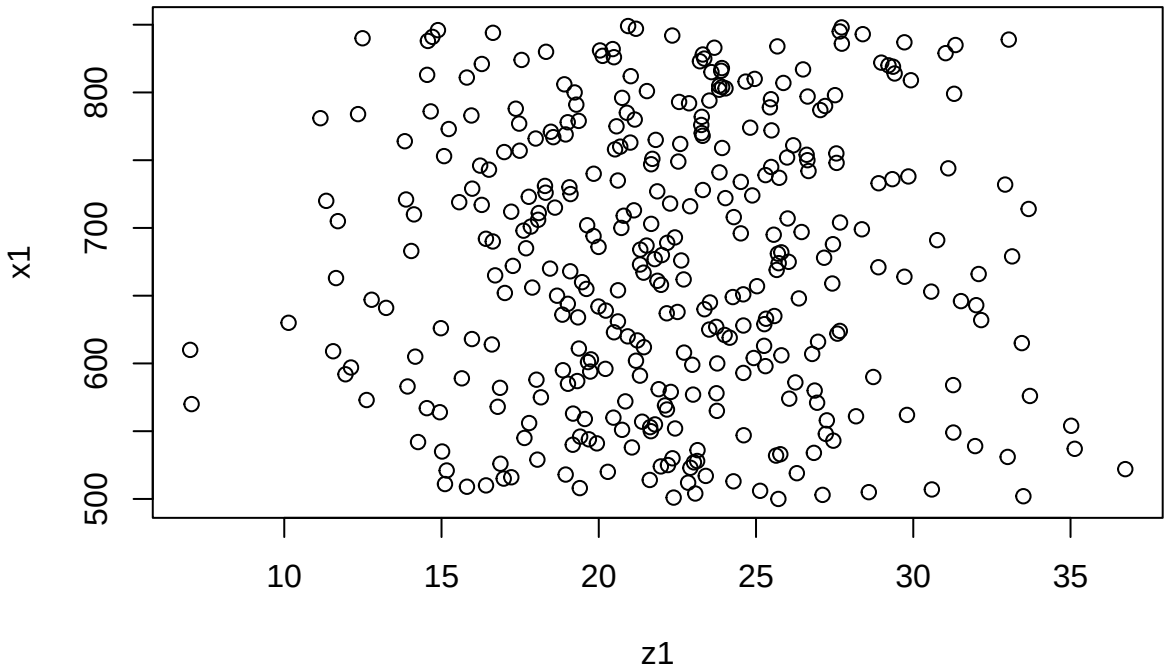

```r
w1<-length(z1)
w1
```

```
## [1] 350
```

```r
x1<-(500:849)
x1
```

```
##   [1] 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517
##  [19] 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535
##  [37] 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553
##  [55] 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571
##  [73] 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589
##  [91] 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607
## [109] 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625
## [127] 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643
## [145] 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661
## [163] 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679
## [181] 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697
## [199] 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715
## [217] 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733
## [235] 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751
## [253] 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769
## [271] 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787
## [289] 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805
## [307] 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823
## [325] 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841
## [343] 842 843 844 845 846 847 848 849
```

```r
plot(z1,x1)
```




CLASE 2

## Ejercicio 1 de algoritmos

Consigna: Las ultimas 3 cifras de mi DNI son 458,consigna crear una variable que tenga ese número

```r
DNI=458
```

Consigna: Crear un vector que tenga todos los números enteros desde el 1 al 458.

```r
Secuencia_dni<- (1:458)
Secuencia_dni
```

```
##   [1]   1   2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18
##  [19]  19  20  21  22  23  24  25  26  27  28  29  30  31  32  33  34  35  36
##  [37]  37  38  39  40  41  42  43  44  45  46  47  48  49  50  51  52  53  54
##  [55]  55  56  57  58  59  60  61  62  63  64  65  66  67  68  69  70  71  72
##  [73]  73  74  75  76  77  78  79  80  81  82  83  84  85  86  87  88  89  90
##  [91]  91  92  93  94  95  96  97  98  99 100 101 102 103 104 105 106 107 108
## [109] 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126
## [127] 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144
## [145] 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162
## [163] 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180
## [181] 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198
## [199] 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216
## [217] 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234
## [235] 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252
## [253] 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270
## [271] 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288
## [289] 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306
## [307] 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324
## [325] 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342
```

```
## [343] 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360
## [361] 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378
## [379] 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396
## [397] 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414
## [415] 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432
## [433] 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450
## [451] 451 452 453 454 455 456 457 458
```

## Ejercicio 2

Calcular la suma de todos los valores del vector Secuencia_dni

```
Total<-0
Valor_final<-length(Secuencia_dni)
for (i in 1:Valor_final) {
Total<-Total+i
}
Total
```

```
## [1] 105111
```

## Ejercicio 3

Repetir el ejerciio anterior pero en Python

```
suma = 0

for i in range(1, 448):
    suma += i

print(suma)
```

```
## 100128
```

## Ejercicio 4

¿Cuánto tarda en correr el código del ejercicio 2?

```
inicio<-Sys.time()
Total<-0
Valor_final<-1000000*length(Secuencia_dni)
for (i in 1:Valor_final) {
Total<-Total+i
}
Total
```

```
## [1] 1.04882e+17
```

```
final<-Sys.time()
final-inicio
```

```
## Time difference of 7.047384 secs
```

## Ejercicio 5

Repetir el ejercicio anterior utilizando la cabeza de Gauss

```r
Valor_final <- length(Secuencia_dni)
Total <- Valor_final * (Valor_final + 1) / 2
Total
```

```
## [1] 105111
```

### Ejercicio 6 : Construir serie de Fibonacci

```r
n <- 10
fibo <- numeric(n)
fibo[1] <- 0
fibo[2] <- 1
for (i in 3:n) {
  fibo[i] <- fibo[i-1] + fibo[i-2]
}
fibo
```

```
##  [1]  0  1  1  2  3  5  8 13 21 34
```

### Ejercicio 7 : Definición matemática recurrente

```r
fibo <- c(0, 1)

for (i in 2:6) {
  fibo[i+1] <- fibo[i] + fibo[i-1]
  print(fibo)
  print(length(fibo))
}
```

```
## [1] 0 1 1
## [1] 3
## [1] 0 1 1 2
## [1] 4
## [1] 0 1 1 2 3
## [1] 5
## [1] 0 1 1 2 3 5
## [1] 6
## [1] 0 1 1 2 3 5 8
## [1] 7
```

### Ejercicio 8: La penitencia de Newton

Subido en un archivo por separado

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```

```
##      speed           dist
##  Min.   : 4.0   Min.   :  2.00
```

```
##  1st Qu.:12.0   1st Qu.: 26.00
##  Median :15.0   Median : 36.00
##  Mean   :15.4   Mean   : 42.98
##  3rd Qu.:19.0   3rd Qu.: 56.00
##  Max.   :25.0   Max.   :120.00
```

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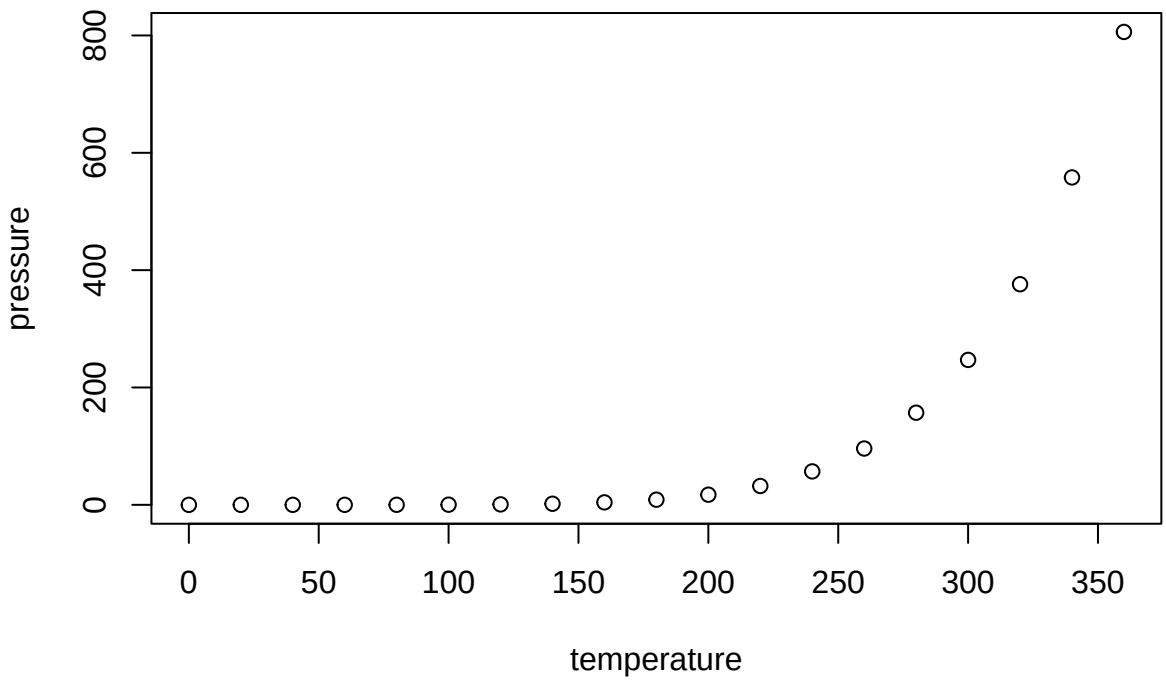


Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.